

Table of Contents

- [SerialCaptureToolAbout.1.02](#)
- [SerialCaptureToolAbout.1.03](#)
- [SerialCaptureToolAbout.1.04](#)
- [SerialCaptureToolAbout.1.05](#)
- [SerialCaptureToolAbout.1.06](#)
- [SerialCaptureToolAbout.1.07](#)
- [SerialCaptureToolAbout.1.08](#)
- [SerialCaptureToolAbout.1.09](#)
- [SerialCaptureToolAbout.1.10](#)
- [SerialCaptureToolAbout.1.11](#)
- [SerialCaptureToolAbout.1.12](#)
- [SerialCaptureToolAbout.1.13](#)
- [SerialCaptureToolAbout.1.14](#)
- [SerialCaptureToolAbout.1.15](#)
- [SerialCaptureToolAbout.1.16](#)
- [SerialCaptureToolAbout.1.17](#)
- [SerialCaptureToolAbout.1.18](#)
- [SerialCaptureToolAbout.1.19](#)
- [SerialCaptureToolDefTopic](#)

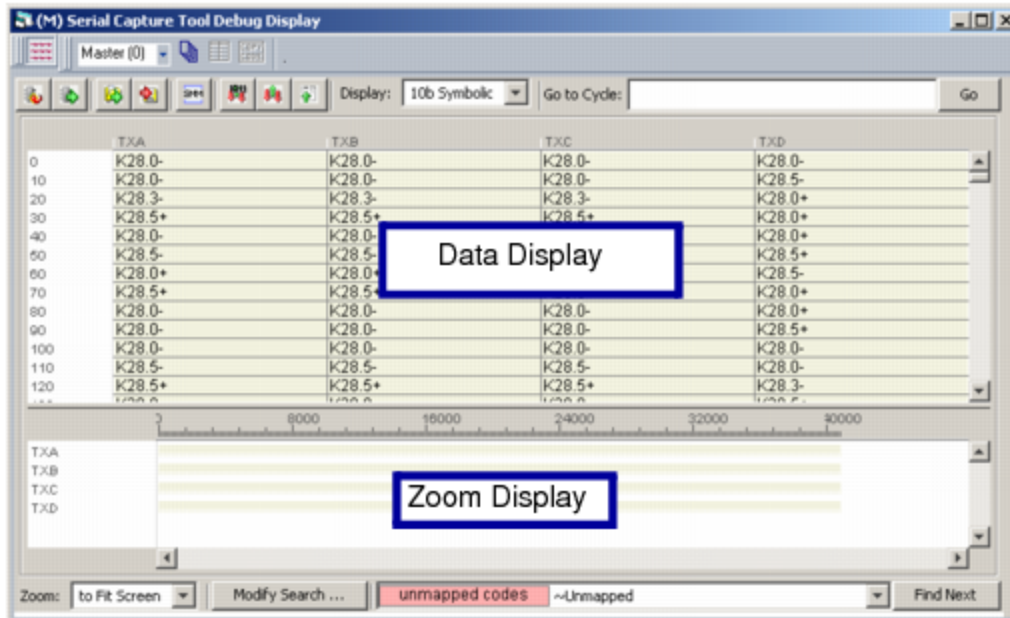
SerialCaptureToolAbout.1.02

	
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Serial Capture Tool : Main Display

Main Display

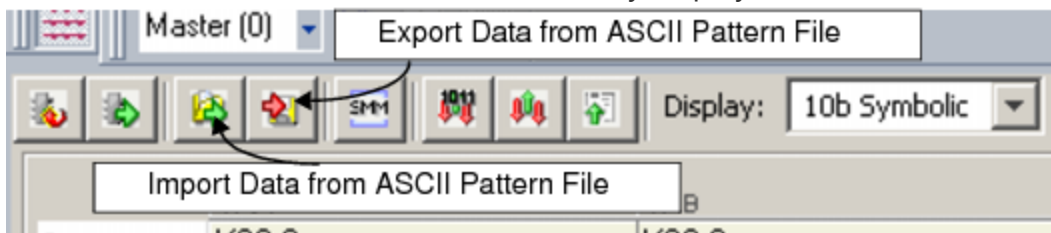
The main display includes two major sections, the data display and the zoom display. The data display displays the decoded serial data in a vertical cycle order. The zoom display displays this same data in a horizontal cycle order.



Serial Capture Tool: Main Display

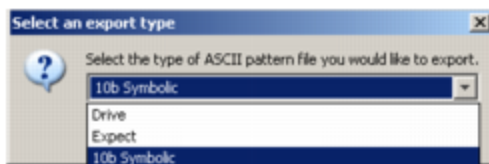
Loading and Saving Serial ASCII Pattern Files

Load data from a serial ASCII Pattern File by clicking the Import Data icon in the main display. After you select a file from the file chooser, the file is scanned for the number of pins and cycles it contains. Once the file is scanned, a summary display is shown.



Import Data and Export Data Icons

The summary display allows you to confirm the information in the ASCII pattern file. You can save data to a file by clicking the Export Data icon in the main display.



Exporting Pattern Data

The Capture Tool can export to one of the following serial pattern file formats:

Drive	A serial ASCII drive pattern file.
Expect	A serial ASCII receive pattern file.

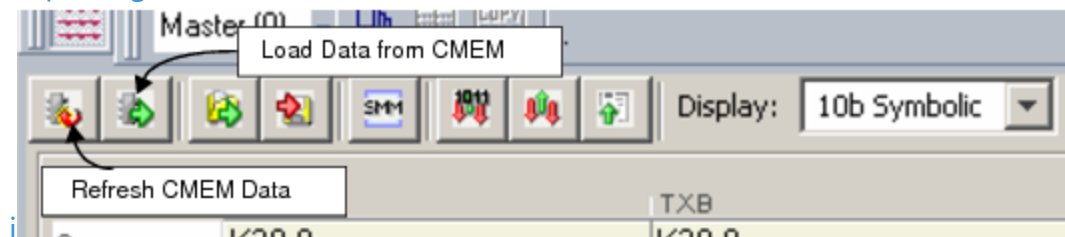
If you select one of these options, give the exported file the .atp extension. You can then import the .atp file, as described above.

The drop-down also includes the option of exporting in whatever display format you have currently selected. For example, if you have 10b XAU selected as the display format, the drop-down contains 10b XAU in addition to Drive and Expect.

Note:

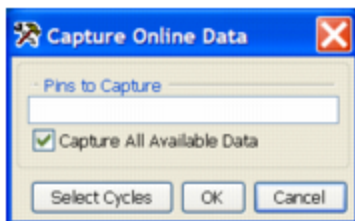
The Serial Capture Tool cannot import data that has previously been exported as 8B Symbolic or 10B Symbolic format,

Capturing Data from CMEM



Capturing Data from CMEM

You can load or refresh data from SB6G or UltraSerial10G capture memory (CMEM). If you select refresh and the Capture Tool is currently empty, you are prompted to specify the data to read back from CMEM.

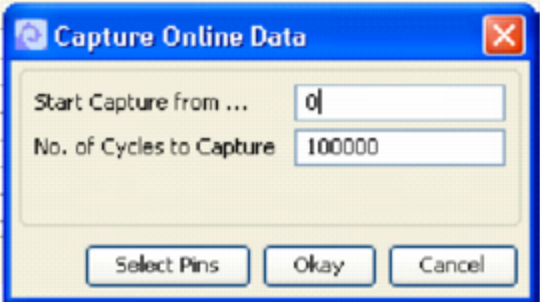


Capture Online Data Dialog Box (Pins View)

The Capture Online Data dialog imports data from CMEM into the Capture Tool.

Pins to Capture	The pins that the Capture Tool should import. Multiple pins should be space delimited (as in, "pin1 pin2 pin3"). Invalid pins are identified, and you are prompted to correct or remove the invalid pins from the list.
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Capture All Available Data Reads back all valid data from CMEM for the specified pins. If this box is not selected, Select Cycles is used to specify a data capture region.



Capture Online Data Dialog Box (Cycles View)

For a specific data capture region, you specify:

- The CMEM cycle to start reading back data from
- The number of CMEM cycles to read back beginning with the Start Capture from cycle

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SerialCaptureToolAbout.1.03

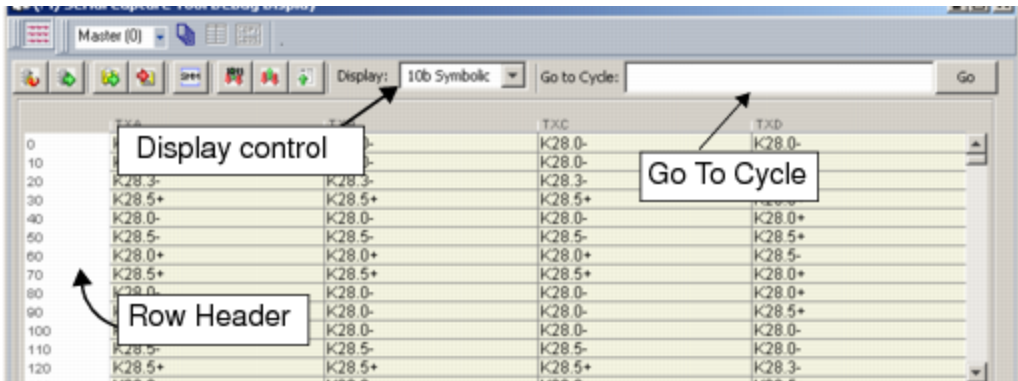
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Serial Capture Tool : Data Display

Data Display

The Data Display shows serialized pin data in decoded 10-bit packets per row in horizontal cycle order. The Display control allows you to select whether you want to view the data as Binary, 8b Symbolic, 10b Symbolic, 10b PCI Exp, or 10b XAUI. For PCI Express and XAUI, the command codes are interpreted.



Data Display Control

Go To Cycle displays the specified cycle in the Data Display and Zoom Display. The row header displays the start cycle number of the decoded 10-bit value. If Zoom is set to Icons, clicking the row header scrolls the Zoom Display to the selected cycle. (See Zoom Display.)

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SerialCaptureToolAbout.1.04

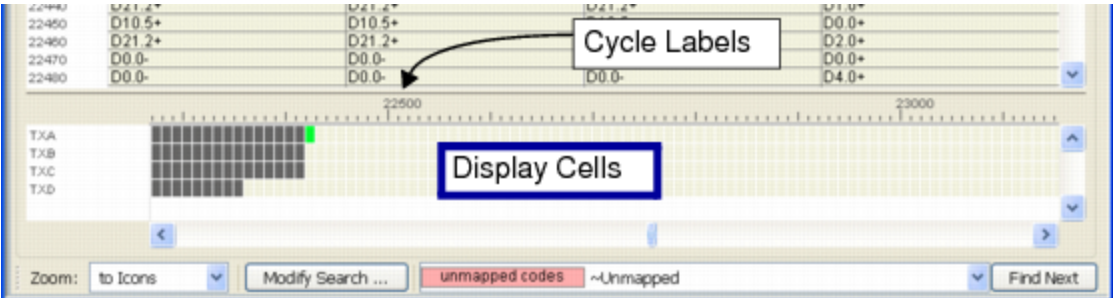
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Serial Capture Tool : Zoom Display

Zoom Display

The Zoom Display provides a quick overview of the captured data. It includes multiple zoom modes so you can zoom from the individual bits of the captured data all the way up to “fit-to-screen” mode.



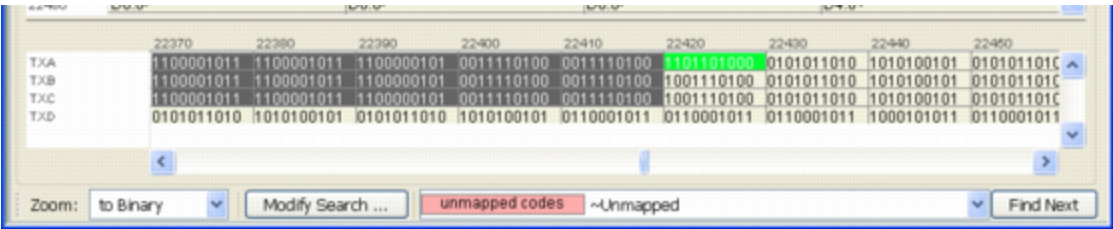
Zoom Display

Display Cells show the data based on the selected zoom mode. Clicking in the display cells region scrolls the Data Display to the cycle selected.

Cycle Labels allow you to get a general idea of the cycle number of each Display Cell.

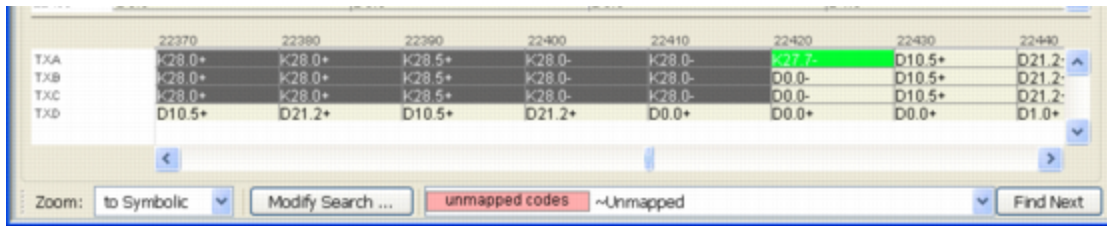
The Zoom selector allows you to select the different zoom modes for this display. Zoom modes are:

- [to Binary](#) provides a view of captured bits framed into 10-bit chunks.



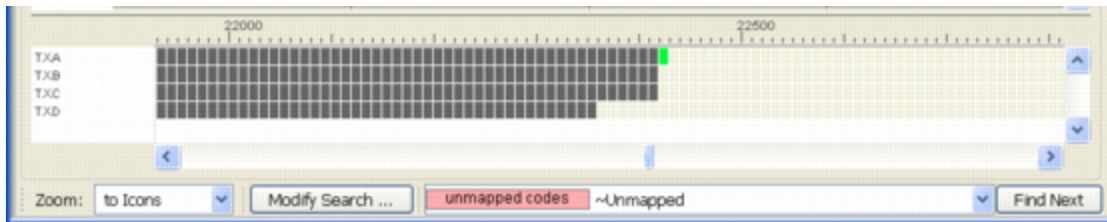
Binary Zoom Display

- [to Symbolic](#) provides a view of the captured data framed in 10-bit chunks and decoded into the 8b/10b IEEE encoding specified by the display modes selector in the Data Display.



Symbolic Zoom Display

- [to Icons](#) provides a view of the data using only the resulting colors from an applied Search Map.



Icons Zoom Display

- [1x View](#) compresses cells to provide four times the number of symbols as the Icons display.



1x Zoom Display

- [16x View](#) compresses cells to provide 64 times the number of symbols as the Icons display.



16x Zoom Display

- [to Fit Screen](#) provides a view of all of the data at once on the screen.



Fit Screen Zoom Display

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SerialCaptureToolAbout.1.05

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Serial Capture Tool : Search Maps

Search Maps

A Search Map provides a set of searches with a name and color that is used to colorize the Data Display and Zoom Display. These searches help you visualize the sequence of data symbols and special symbols in the data set. They are configurable and consist of a number of user specified search parameters. The search map provides a way to easily find the next occurrence of a search parameter in the Data Display and Zoom Display. In addition, you can save and load the search map from disk.

This section describes the following topics:

- | | |
|---|--|
| • | Creating or Modifying Searches |
| • | Adding Searches |
| • | Deleting All Searches |
| • | Loading Search Maps |
| • | Saving Search Maps |
| • | Search Parameters |
| • | Reordering Searches |

- [Enabling Searches](#)
- [Editing Searches](#)
- [Deleting Searches](#)
- [Applying Searches](#)

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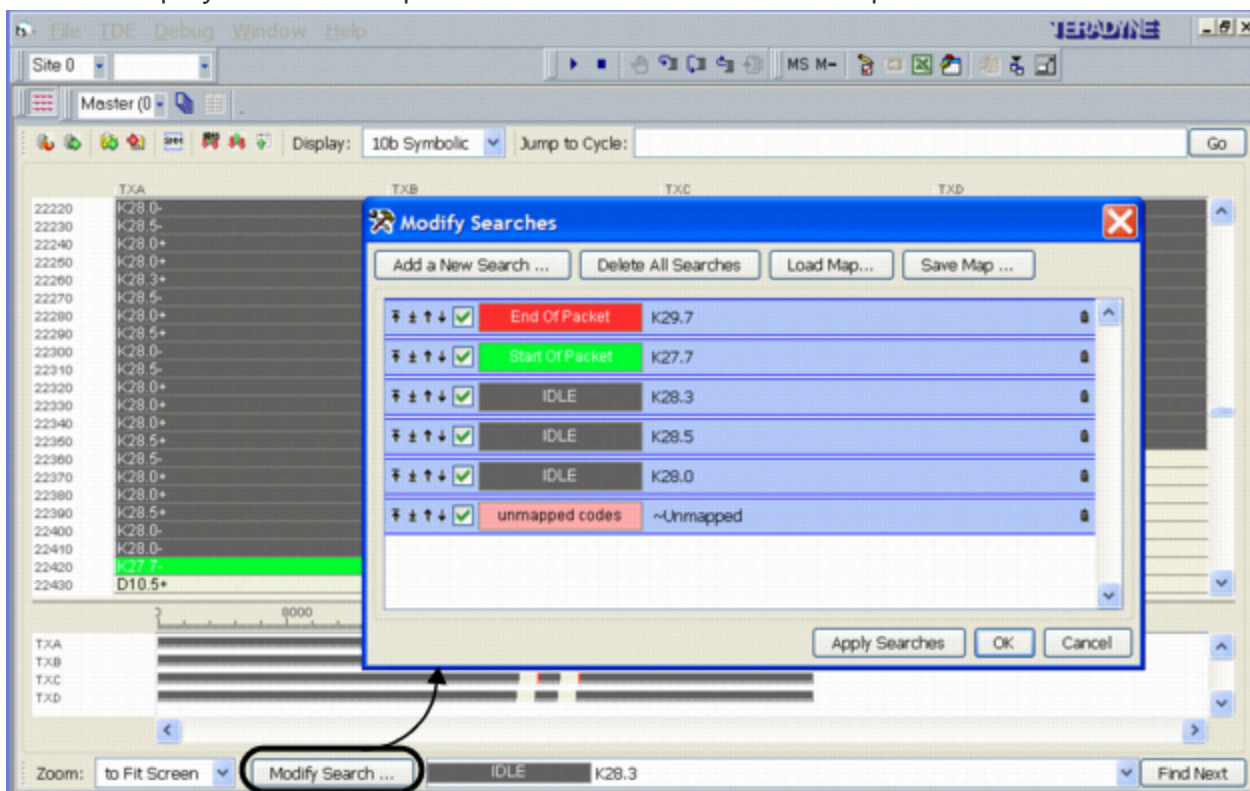
SerialCaptureToolAbout.1.06



Serial Capture Tool : Search Maps : Creating or Modifying Searches

Creating or Modifying Searches

To create a new search or modify an existing search, click the [Modify Search](#) button in the Zoom Toolbar display. This button provides access to the Search Map window.



Search Map Window

The Search Map window allows you to add a new search, delete a search, load an existing search map, and save the current search map. The window shows all available search parameters included in the current search map. The default search parameter is unmapped codes.

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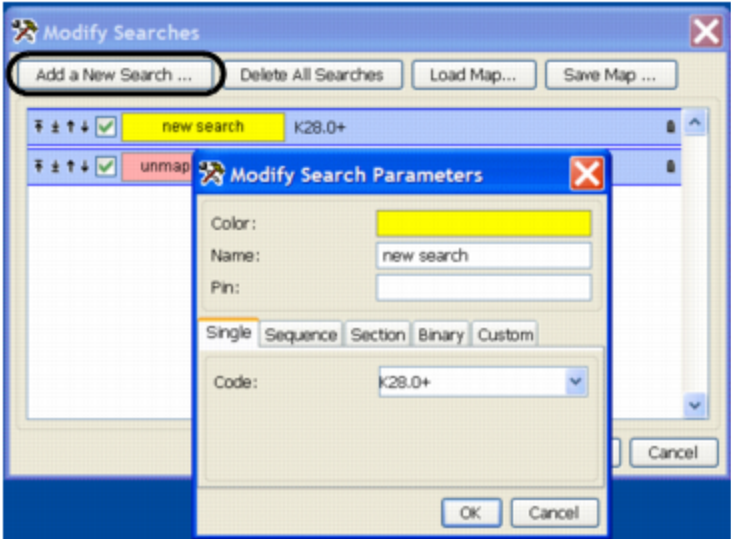
SerialCaptureToolAbout.1.07

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Serial Capture Tool : [Search Maps](#) : Adding Searches

Adding Searches

Clicking the Adding a New Search button brings up the Modify Search Parameter window which inserts a new search parameter into the map. Set the parameters as specified in [Search Parameters](#). Once finished, click the OK button to apply or the Cancel button to cancel the search.



Adding a New Search

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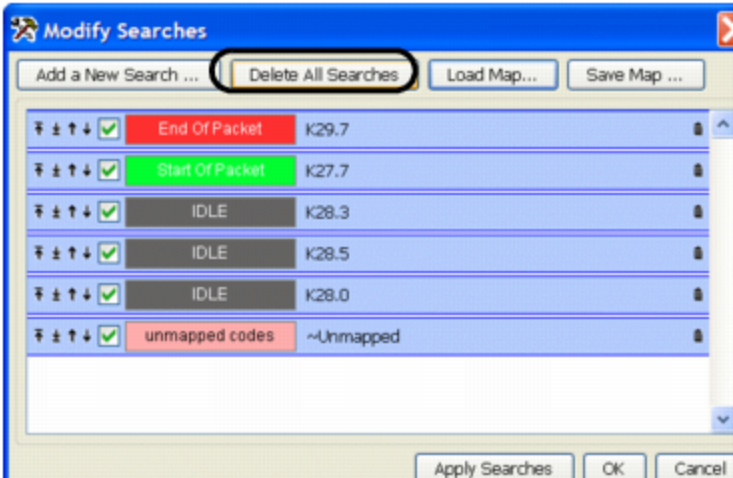
SerialCaptureToolAbout.1.08

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Serial Capture Tool : Search Maps : Deleting All Searches

Deleting All Searches
Delete All Searches removes every search parameter in this map except for the Unmapped Codes search.



Deleting All Searches

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SerialCaptureToolAbout.1.09

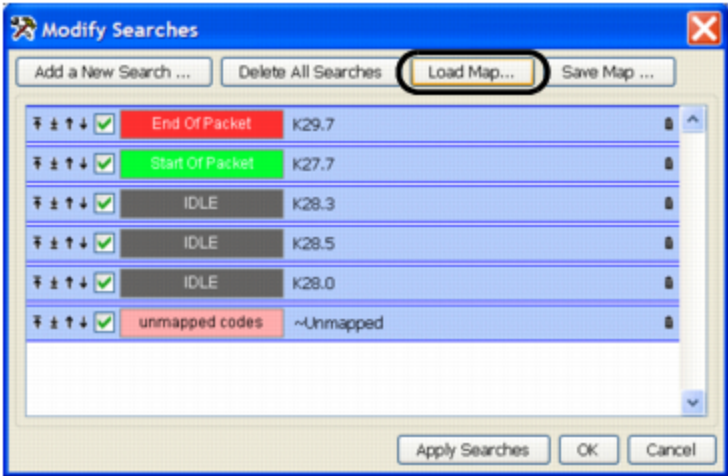
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Serial Capture Tool : Search Maps : Loading Search Maps

Loading Search Maps

The Load Map button loads the Search Map from a specified file. Once a search map is loaded, you must apply the new search parameters to the data using the Apply Searches button.



Loading a Search Map

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SerialCaptureToolAbout.1.10

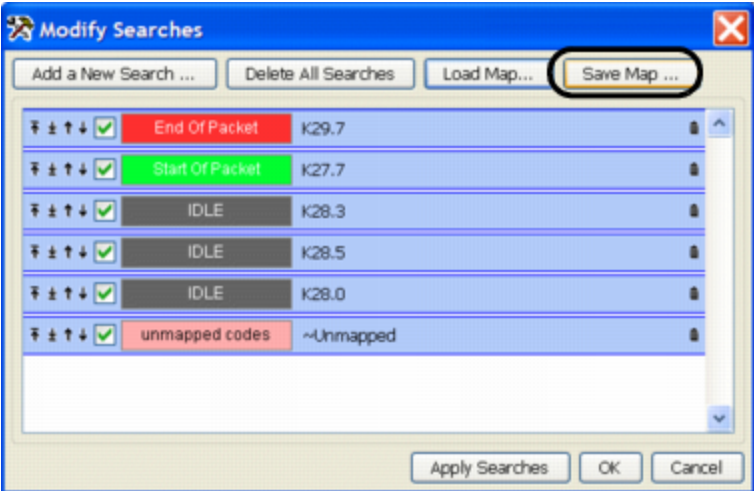
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Serial Capture Tool : Search Maps : Saving Search Maps

Saving Search Maps

The Save Map button saves the Search Map to a user specified file.



Saving a Search Map

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SerialCaptureToolAbout.1.11

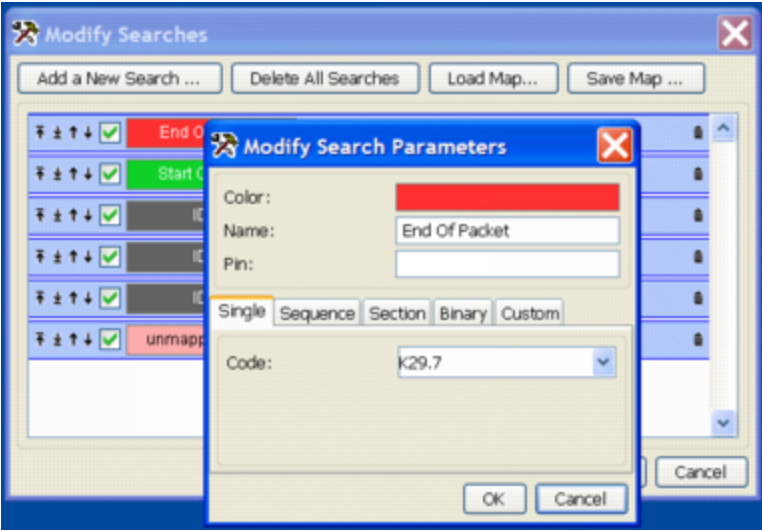
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Serial Capture Tool : Search Maps : Search Parameters

Search Parameters

Search Parameters looks for specific items in the captured data and highlights that data with a specified color. The Modify Search Parameters window allows you to set up parameters for a search.



Setting Up Parameters for a Search

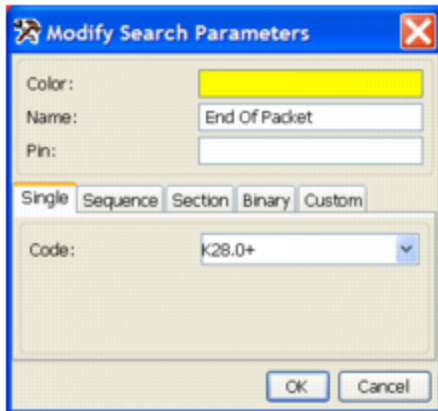
The Modify Search Parameters window is comprised of two main sections. The top section specifies common settings for every type of search parameter. The bottom section contains tabs which allow you to set up one of several search types.

Color	The color to be used with the current search.
Name	The name of the current search.
Pin	The pins to apply the search to. The default is blank, which means the search applies to all pins currently being displayed in the Capture Tool.

The following section describes each type of search and its associated parameters.

Single Search

Single search looks for a particular IEEE 8b/10b code within the data.



Setting Up a Single Code Search

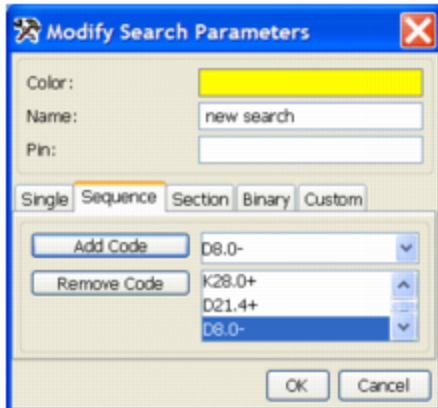
In this example, any K28.0+ code that appears in the data will be colored yellow. The drop-down selector for code is editable, so you can type in a particular code you want to find.

✓ **Note:**

If you omit the + or - sign from the code, then the search finds the token with either polarity.

Sequence Search

Sequence search looks for a sequence of IEEE 8b/10b codes in the data. The Sequence Search scans the data for the entire sequence and only colors the data if every code matches in the proper order.



Setting Up a Sequence Search

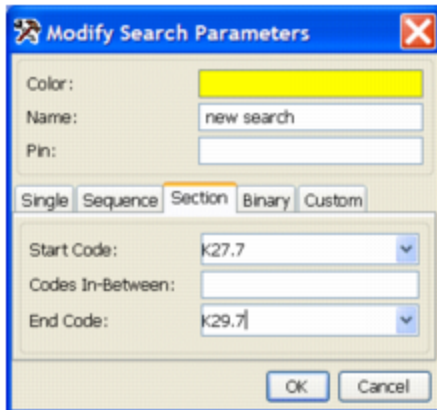
In this example, any K28.0+ code followed by a D21.4+ code and a D8.0- code, will be colored yellow. There is no limit to the length of search sequences. The drop-down selector is editable, so you can type in a particular code you want to find.

✓ **Note:**

If you omit the + or - sign from the code, then the search finds the token with either polarity.

Section Search

Section search looks for a section of codes delineated by a start and end code. If specified, a section can also contain a specific number of codes in between the start and end codes.



Setting Up a Section Search

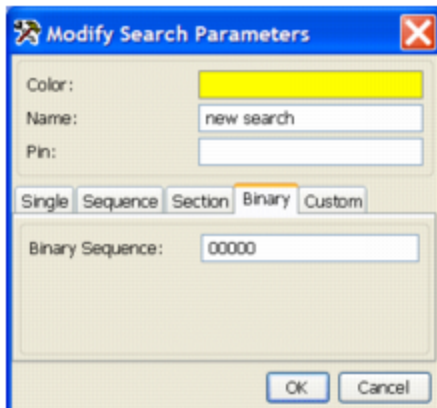
In this example, any sequence that starts with K27.7 code and ends with K29.7 code will be colored yellow. Here Codes In-Between is left blank, so any number of codes can occur between the start and end code. If Codes In-Between is specified, the number of codes between the start and end code must match exactly. You can edit the drop-down selector, so you can type in a particular code you want to find.

☒ **Note:**

If you omit the + or - sign from the code, then the search finds the token with either polarity.

Binary Search

Binary search looks for a particular sequence of binary bits. This search is unique in that it is not bounded by the 10-bit frame used for 8b/10b decoding.



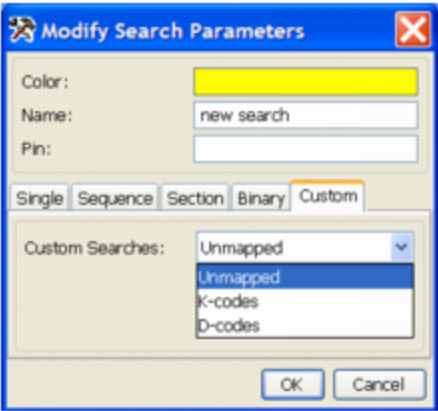
Setting Up a Binary Search

In this example, if five zeroes span two 10-bit codes, those two 10b codes will be highlighted with yellow.

There is a 63 bit limit to the binary search sequence. Any binary sequence longer than 63 bits is truncated at 63 bits.

Custom Searches

Custom searches are special searches that do not have any user specified parameters.



Setting Up a Custom Search

Custom searches include:

Unmapped	Looks for 10b codes that do not map to any known IEEE 8b/10b codes.
K-codes	Looks for any IEEE 8b/10b command codes.
D-codes	Looks for any IEEE 8b/10b data codes.

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SerialCaptureToolAbout.1.12



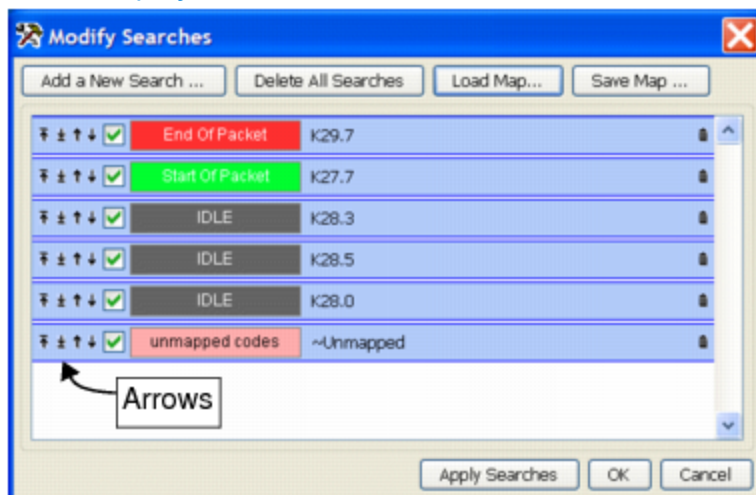
Serial Capture Tool : Search Maps : Reordering Searches

Reordering Searches

Search Order is important when dealing with multiple search parameters. Any matching data is first colored by the top search parameter in the table. Data that was not previously colored is then available to be colored by subsequent searches. Any matching data is then colored by the second search parameter. This colorization process continues until all search parameters have been used to colorize the data.

For example, you can create two searches: one search that colorizes all K codes, and another that colorizes only one K code (say K27.7). In this case, the order the searches appear in the table is important. If you do not search for the single K code first, then the all K code will mask the single K code colorization.

You can reorder the search parameters to prevent unwanted masking from occurring using arrows in the display.



Using Arrows to Reorder Search Parameters

Arrows allow you modify the importance of search parameters. From left-to-right, the arrows are Top, Bottom, Up, and Down. Top sends the selected search parameter to the top of the list. Bottom

sends the selected search parameter to the bottom of the list. The Up and Down arrows shift the selected search parameter one position in the search order.

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SerialCaptureToolAbout.1.13

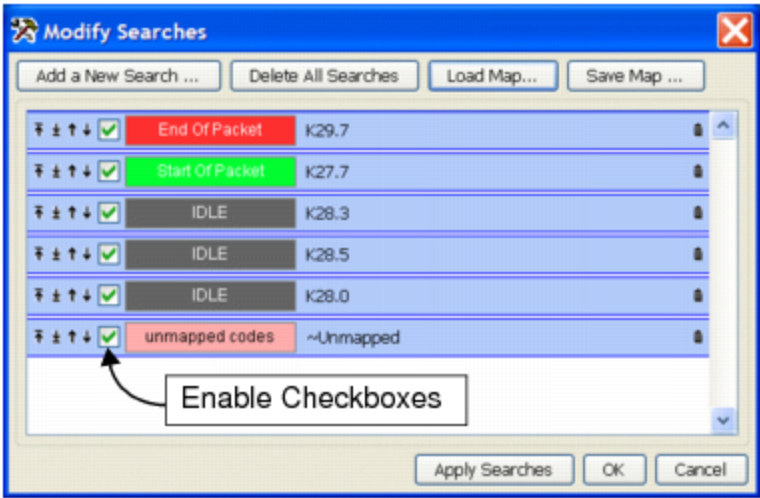
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Serial Capture Tool : Search Maps : Enabling Searches

Enabling Searches

The Enable Checkbox determines whether a search parameter is used to colorize the data.



Enable Checkboxes

If the Enable Checkbox is checked, the search parameter is used to colorize the data. All new search parameters are enabled by default.

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SerialCaptureToolAbout.1.14

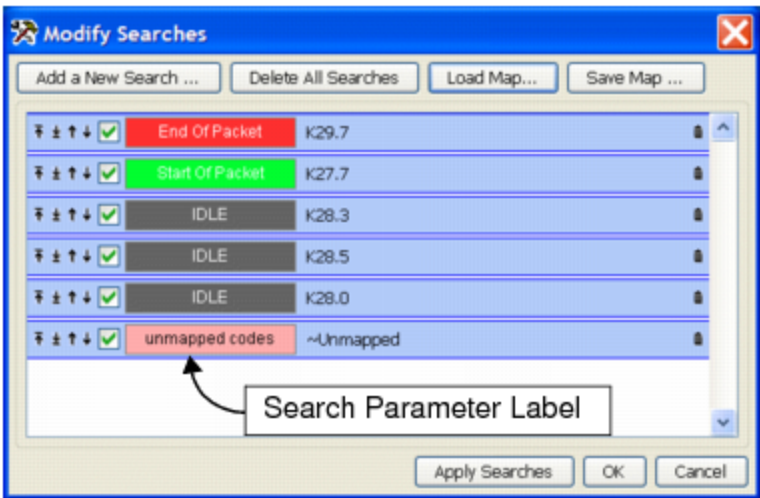
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Serial Capture Tool : Search Maps : Editing Searches

Editing Searches

Once you create a search, you can select an existing search parameter to bring up the Modify Search Parameter window for editing.



Editing an Existing Search Parameter

Modify the parameters as specified in Search Parameters. Once finished, click the OK button to apply or the Cancel button to cancel the search.

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SerialCaptureToolAbout.1.15

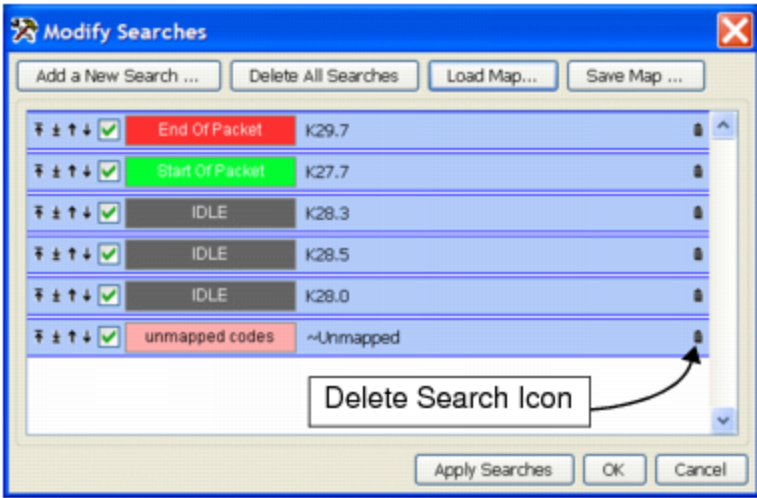
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Serial Capture Tool : [Search Maps](#) : Deleting Searches

Deleting Searches

The Delete Search Icon deletes the specified search parameter.



Deleting an Individual Search

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SerialCaptureToolAbout.1.16

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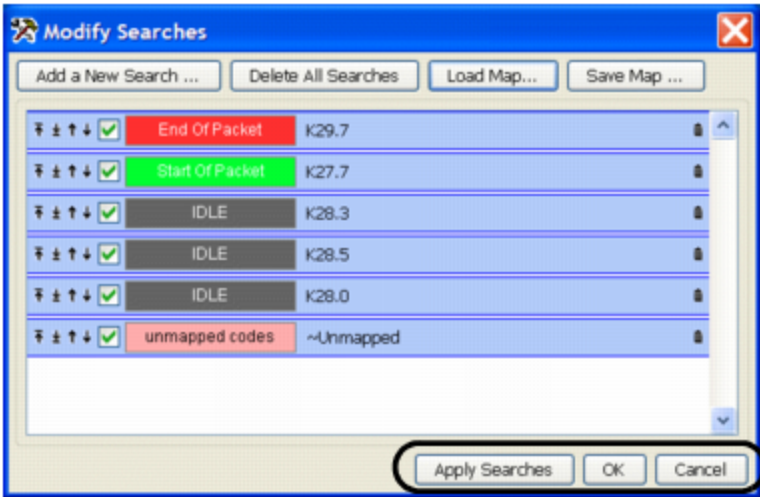
>

Serial Capture Tool : Search Maps : Applying Searches

Applying Searches

Once you create the desired set of searches, you can use the Apply Searches or OK button to colorize the data.

Apply Searches	Colorizes the data using the existing search parameters.
OK	Applies the search and closes the Modify Searches window.
Cancel	Cancels any changes made to the existing search map and closes the Modify Searches window.



Applying Searches

☒ **Note:**

The currently loaded search map is applied to the current and any future loaded or captured serial data.

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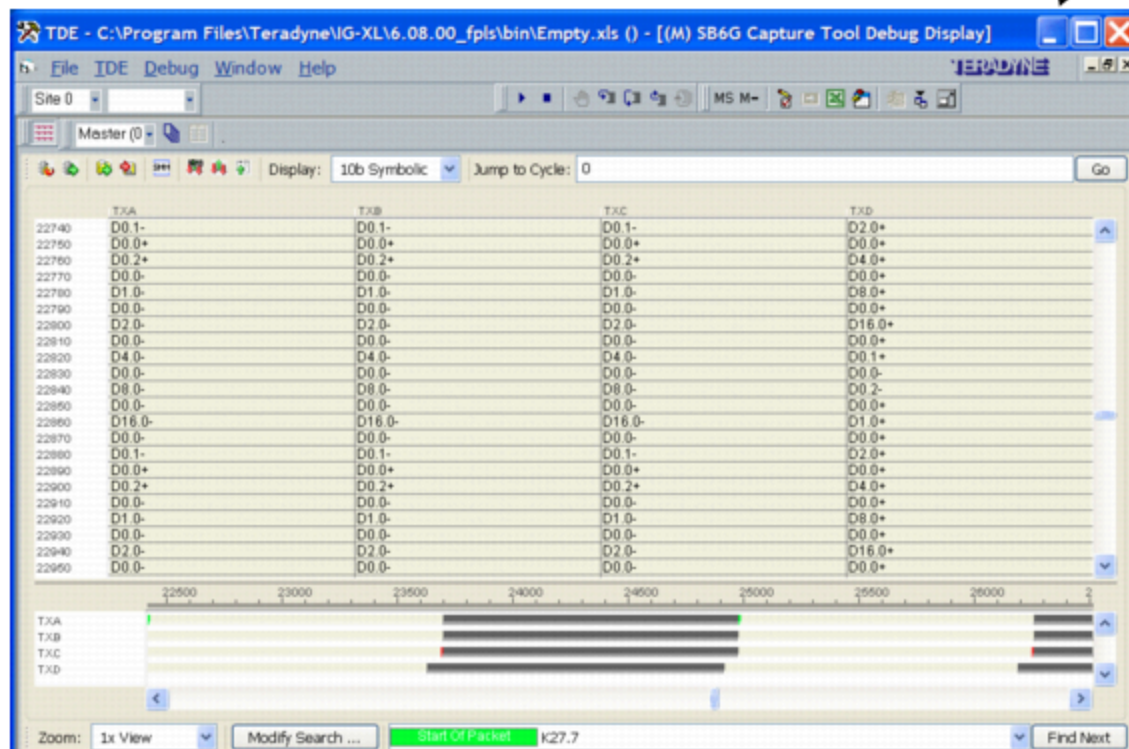
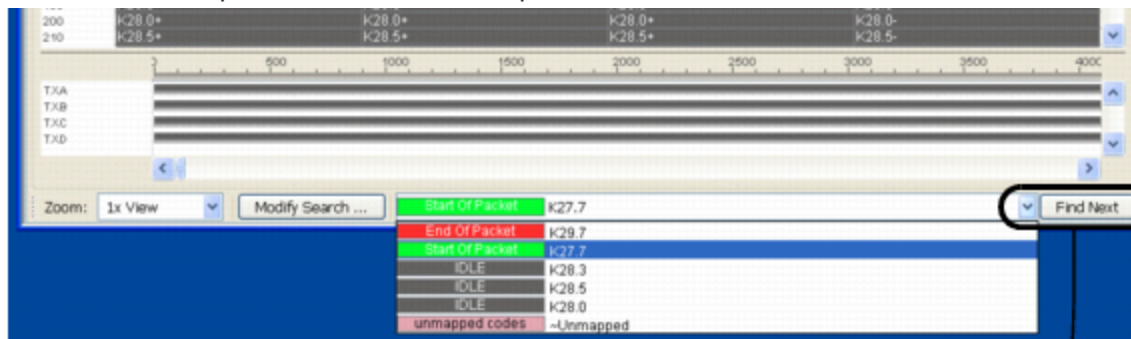
SerialCaptureToolAbout.1.17



Serial Capture Tool : Finding the Captured Data

Finding the Captured Data

Find Next allows you to quickly find items within the captured data. You can select a previously defined search parameter or enter a parameter to use for the search.



Using the Find Next Button

In this example, the Search Map contains a search parameter called "Start Of Packet" that finds all instances of the K27.7 code and colors them green. When the Find Next button is selected, the Data Display and Zoom Display are scrolled to the next K27.7 code in the captured data. In addition to searching for predefined search parameters, you can also enter a particular IEEE code in the text area of the search selector. When you click the Find Next button, the next occurrence of the code is selected and colored. These items are colored yellow by default.

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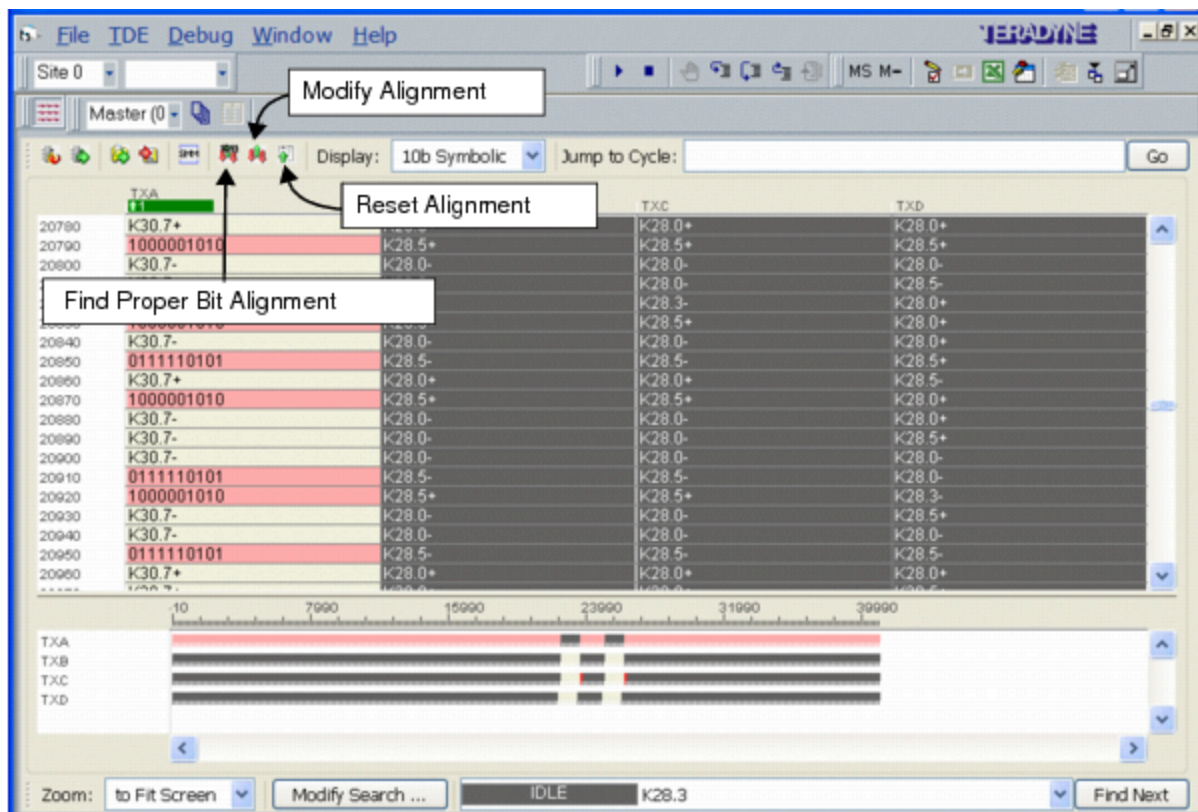
SerialCaptureToolAbout.1.18



Serial Capture Tool : Modifying Alignment

Modifying Alignment

Captured data may be out of alignment by several bits or by 10-bit cycles. The Capture Tool can shift the data by one or more cycles. When the data is shifted, the start of the data begins at a different bit.



Alignment Icons

In this example (due to the large number of unmapped codes in TXA), we can visually determine there is a bit alignment issue in that pin.

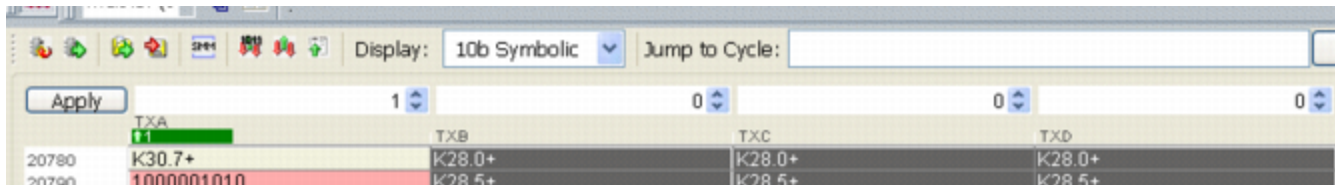
Finding Bit Alignment

[Find Bit Alignment](#) shows the Find the Correct Bit Alignment window, which allows you to automatically find the best bit alignment per pin.

[When you select the Find Bit Alignment icon](#), the tool scans the data and determines the best alignment per pin. When the scan is complete, the Find the Correct Bit Alignment window appears. The Find the Correct Bit Alignment window indicates either: "Data appears to be misaligned" or "Unable to find a better bit alignment." To apply the determined alignment, click the Apply Alignment button.

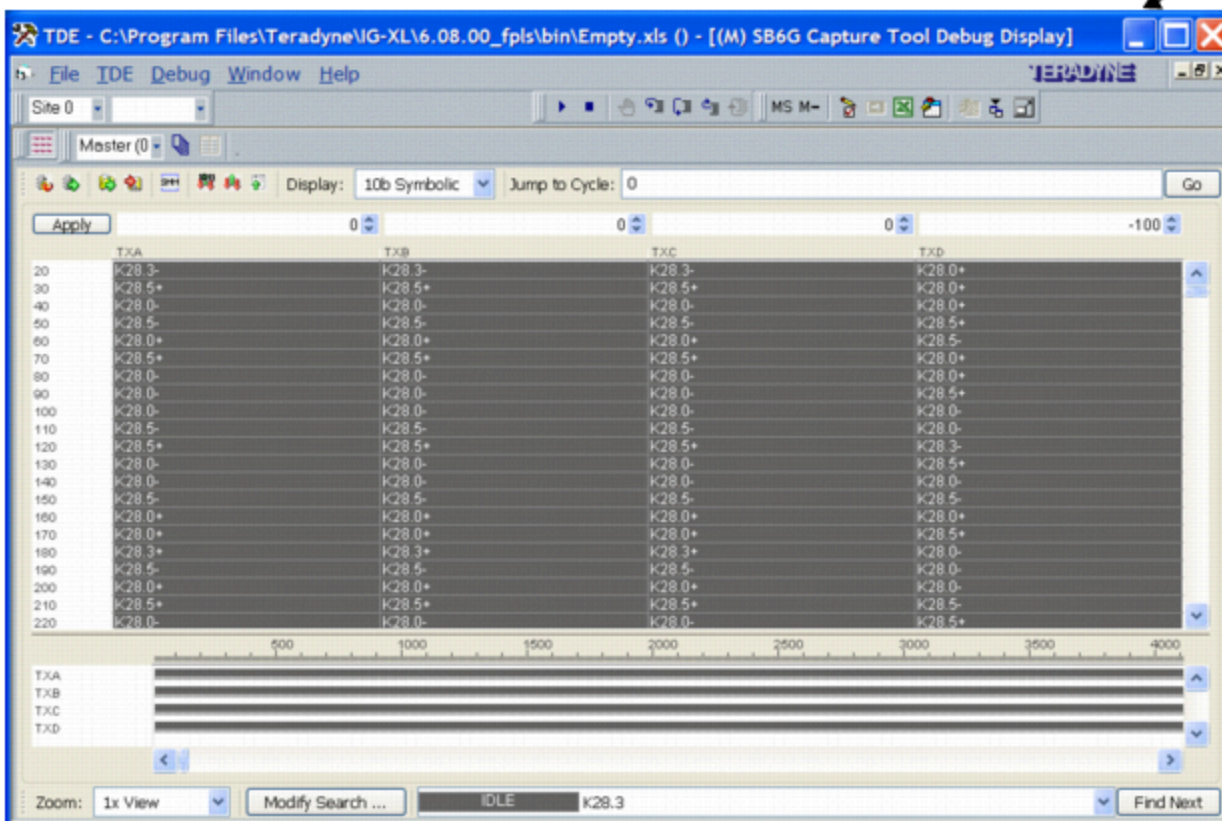
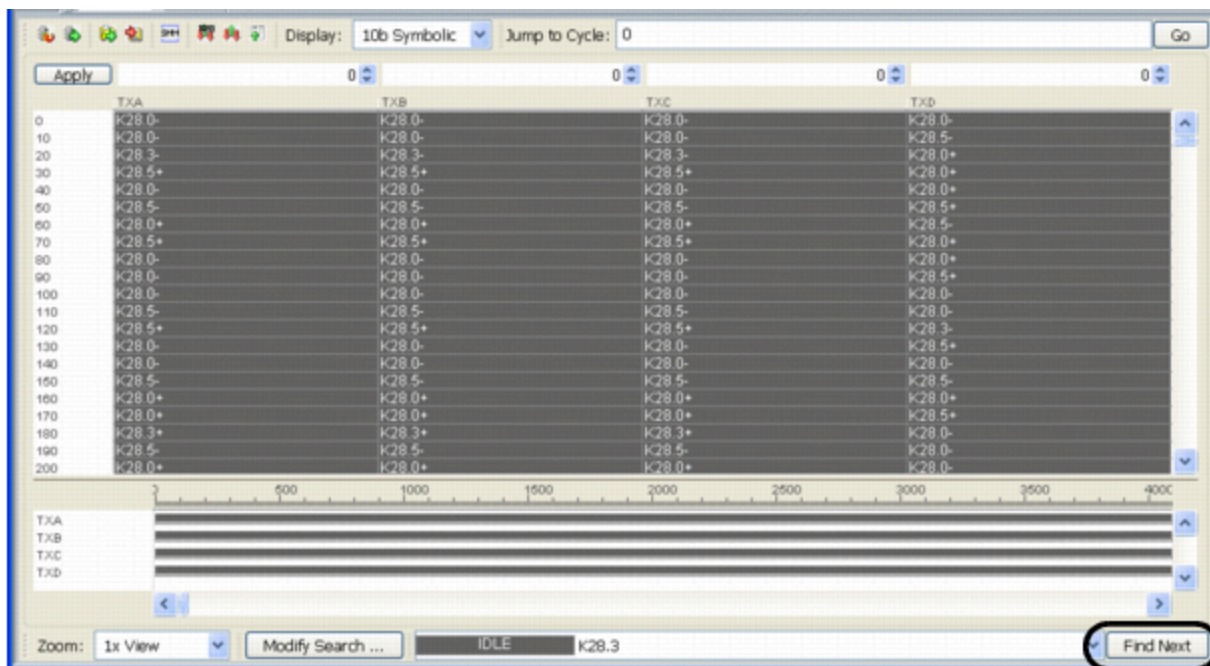
[Modifying Alignment](#)

[Modify Alignment](#) shows and hides the alignment window, allowing you to modify the alignment per pin.



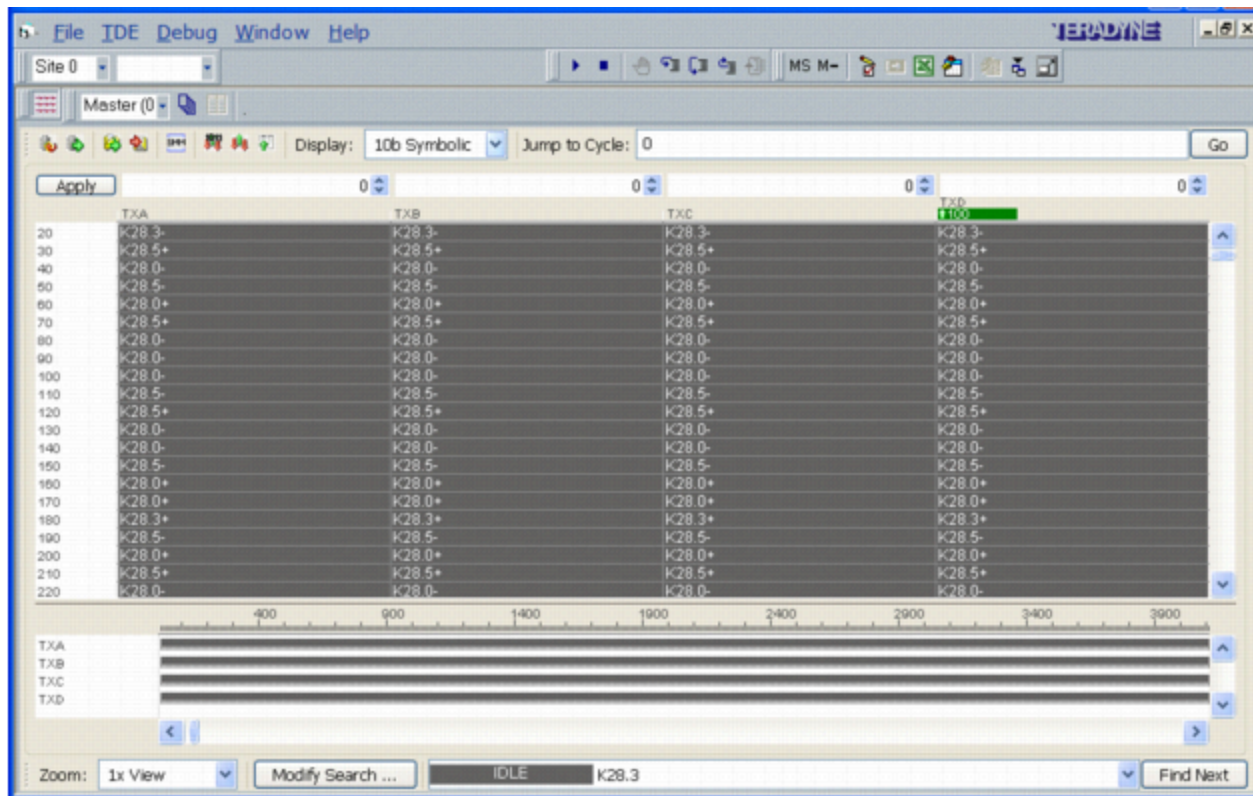
[Alignment Window](#)

The alignment window enables manual cycle shift adjustments. Each pin has a text box that specifies how much to shift the data. Clicking the up or down arrows at the end of the text box shifts the data a standard amount of 10 cycles (equivalent to a 10b code). You can enter in any specific value. A positive value shifts the data down, and a negative value shifts the data up. When data are shifted, a green or red box appears indicating the amount of movement from the original position. A green box indicates the data has shifted up by the specified number of cycles. A red box indicates the data has shifted down by the specified number of cycles. Entering a new shift value moves the data relative to the existing shift value. To return to the original alignment for a pin, enter an equal and opposite value to the current shift value. In addition to manual cycle shifts, you can also shift the data based on a particular 10b code.



Using the Find Next Button

Select a search parameter and use the Find Next button find the next occurrence. The search parameter for all pins may not occur in the same cycle. In this case, the alignment window displays the shift values that will make all pins align to the search parameter.

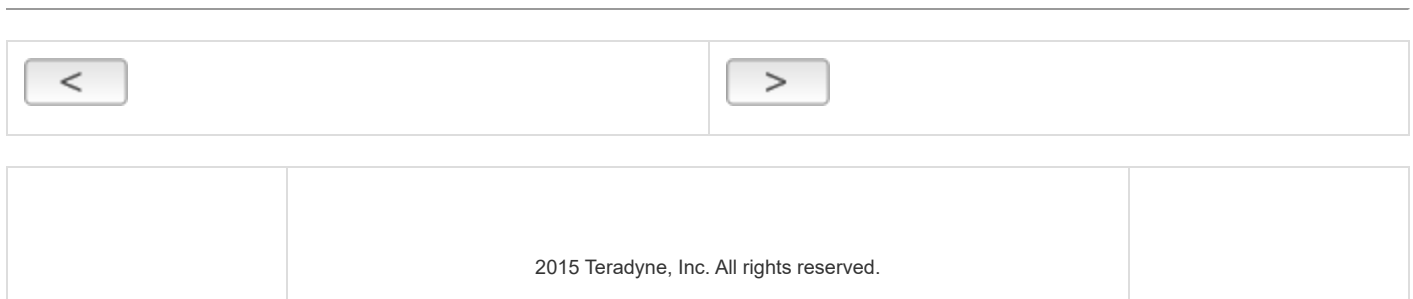


Applying Bit Alignment

Click the Apply button to make all pins align to the search parameter.

Resetting Alignment

Reset Alignment cancels any alignment modifications and restores the original data set for all pins.



SerialCaptureToolAbout.1.19



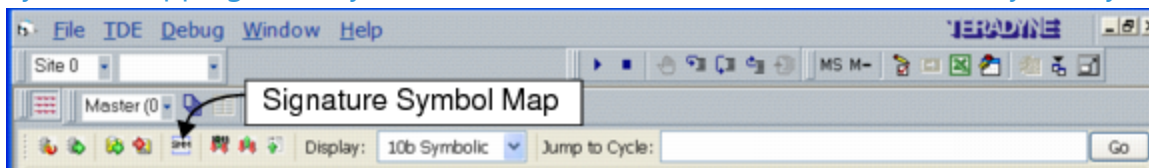
Serial Capture Tool : Symbol Mapping

Symbol Mapping

✓ **Note:**

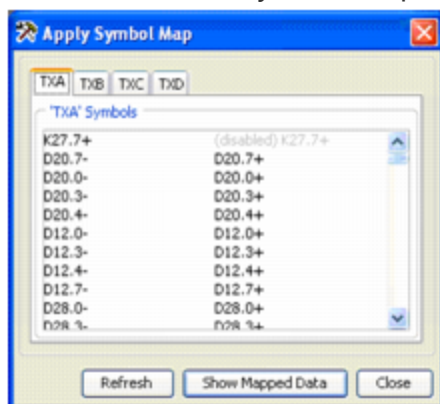
Symbol mapping does not apply to the UltraSerial10G because this instrument does not support signature receive.

Symbol mapping allows you to view the data after it has been filtered by the Symbol Map Memory.



Signature Symbol Map Icon

The Signature Symbol Map icon brings up the Apply Symbol Map window, which allows you to view the current Symbol Map and filter the data.



Apply Symbol Map Window

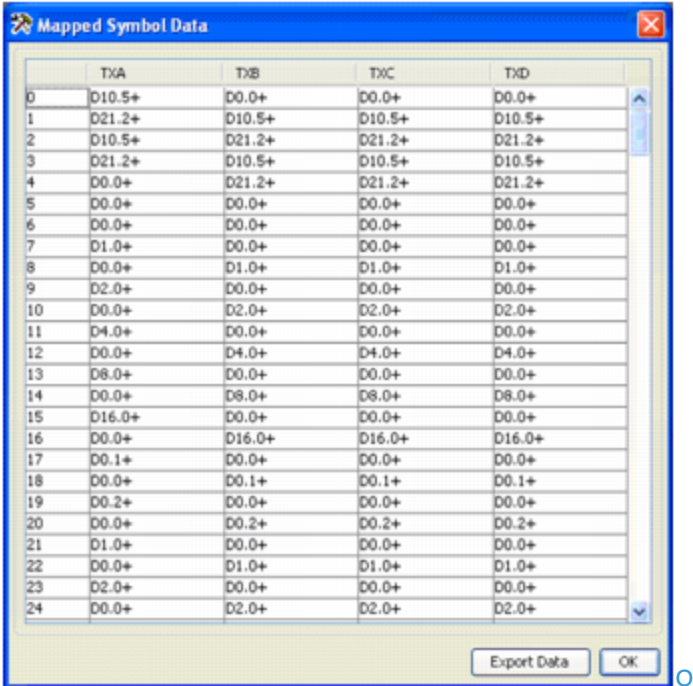
The Apply Symbol Map window displays any symbol that you have mapped to another symbol or disabled on a per-pin basis. The window omits symbols that are set to the default (one-to-one mapping).

✓ **Note:**

In the IEEE 8b10b encoding standard, a number of data symbols have the same 10b representation for both positive and negative disparity. In these cases, the code is always be interpreted as the positive disparity symbol. For example, the 10b representation for symbol D20.1+ (positive disparity) is "001011 1001". The 10b representation for symbol D20.1- (negative disparity) is also "001011 1001". Therefore, if you map D20.1- to D20.1+, the Apply Symbol Map window will not show that D20.1- was mapped.

The Refresh button updates any changes that were made to Symbol Map Memory using VBT language.

The Show Mapped Data button applies the Symbol Map Memory to the captured data.



Mapped Symbol Data Window

The Mapped Symbol Data window shows the data after the symbols have been mapped to the specified symbols and disabled symbols have been removed.

The Export Data button saves the filtered data to a file of one of the following types:

Drive	An SB6G ASCII drive pattern file.
Expect	An SB6G ASCII receive pattern file.
10b Symbolic 10b PCI Exp 10b XAUI 8b Symbolic Binary	An ASCII file that contains the data as displayed in the table (based on your Display option). Note that you cannot later import a file that was exported using any of these type.

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SerialCaptureToolDefTopic

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Serial Capture Tool

Serial Capture Tool

The Serial Capture Tool is used to capture and search serialized data stored in SB6G or UltraSerial10G capture memory (CMEM). High speed serial devices output 10-bit words that are constructed from individual bits for a given pin across 10 cycles. The Capture Tool translates these 10-bit words into IEEE 8b/10b encodings and displays them.

Once DUT test results are stored in capture memory, the Capture Tool queries the memory (CMEM) for data and decodes the serialize data into 8b/10b codes. Capture Tool can also read data from a file and export data to a file.

The Capture Tool allows you to:

- Fix data may become misaligned if the capture memory begins the data capture with a misaligned 10-bit data frame. The tool allows you to fix these alignment issues on the display before decoding the words.
- Quickly identify particular 8b/10b codes, sequences of codes, and sections of codes. This tool also identifies 10-bit words that cannot be mapped into an 8b/10b code.
- View the symbol map memory setup.
- View the data after it has been filtered by the symbol map memory. You can view the data that is sent to the signature generator and used to create a signature.

The Serial Capture Tool features are described in the following topics:

•	Main Display
•	Data Display
•	Zoom Display
•	Search Maps
•	Finding the Captured Data
•	Modifying Alignment
•	Symbol Mapping

For information on other debugging tools used with SB6G and UltraSerial10G, see these topics:

•	Jitter Tool
•	SB6G Debug Displays
•	UltraSerial10G Debug Displays